Hybrid Music Recommender



Combining content-based similarity with item-item collaborative filtering.

PREPARED BY NIRMIT GUPTA

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Project Objectives

Building a complete, reproducible hybrid recommender architecture from raw Spotify metadata.

+ TARGET → TOP 10

+

HYBRID ENGINE

Alpha-Weighted Blend

Combine content-based and item-item collaborative filtering scores with a tunable alpha parameter (default 0.5).

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INTERFACE

Interactive Streamlit

Searchable song and genre input modes with transparent scoring (Content, Collaborative, and Hybrid metrics).

COMPUTATION

Sparse Scalability

Efficient sparse matrix operations for user-track interactions, avoiding dense catalog-wide similarity matrices.

EVALUATION

Ranking Metrics

Performance measured via Precision@10, Recall@10, MAP, and NDCG using held-out synthetic events.

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System Architecture

Modular pipeline designed for clean data separation and graceful degradation.

+ SCALABLE · SPARSE

PREPROCESSING

Parquet Storage

- **Cleaning:** De-duplication & imputation
- **Optimization:** Parquet for fast loading
- **Features:** Standardized audio metrics

ENGINES

Content & Collab

- **Content:** TF-IDF Artist + Genre OHE
- **Collab:** Item-Item Sparse Matrix
- **Sim:** Cosine nearest-neighbors

Pipeline: Raw → Parquet → Model → UI

BLENDING

Hybrid Ranker

- **Normalizing:** Min-max per candidate
- **Blending:** Weighted alpha default 0.5
- **Degradation:** Content-only fallback

DEPLOYMENT

Streamlit App

- **Interface:** Searchable song/genre select
- **Persistence:** Joblib model storage
- **Transparency:** Component score display

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Hybrid Methodology

Deep feature engineering combined with sparse behavioral modeling.

+ METADATA → BEHAVIOR

+

CONTENT ENGINE

Feature Pipeline

- **Audio:** Standardized danceability, energy, valence
- **Genre:** One-hot encoding for exact matching
- **Artist:** TF-IDF vectors for stylistic similarity

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COLLABORATIVE ENGINE

Interaction Matrix

- **Sim Interactions:** Reproducible genre-biased histories
- **Item-Item:** Sparse user-track interaction matrix
- **Norm:** Cosine similarity over binary interaction

VALIDATION

Held-out Protocol

- **Test Split:** One random interaction removed per user
- **Training:** Collaborative rebuilt on reduced set
- **Leakage:** Zero data leakage during evaluation

OPTIMIZATION

Hyper-parameters

- **Neighbors:** min(catalog, 301) for search
- **Seed:** Fixed seed (42) for reproducibility
- **Alpha:** Optimized for NDCG@10 via grid search

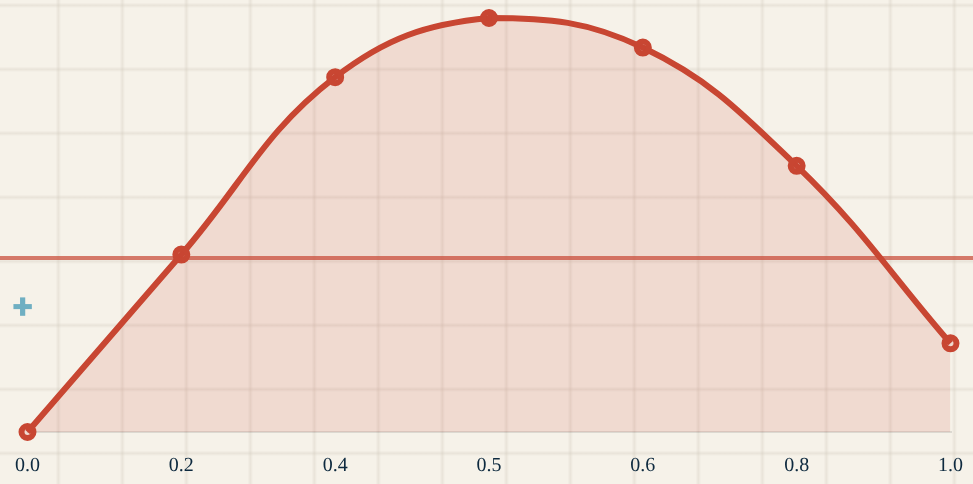
Ranking & Optimization

Score normalization and weighted blending for balanced discovery.

HYBRID FORMULA

$$Score = \alpha \cdot Content + (1 - \alpha) \cdot Collaborative$$

α optimized via grid search on NDCG@10



RANKING CONSTRAINTS

Min-Max Normalization

Raw similarity scores are relative; normalization happens within the top-N candidate set to ensure additive compatibility.

Seed Handling

The input seed track is strictly excluded from results. Result sets are de-duplicated by track ID before final ranking.

Optimized Defaults

SONG MODE A	GENRE MODE A
0.6	0.5

Grid Search

Sweep of α from 0.0 to 1.0 (step 0.1) on 500 synthetic users to maximize NDCG@10.

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Evaluation Results

Performance metrics computed on 200 synthetic users via held-out validation.

+ HYBRID WINS NDCG

ACCURACY

Precision@10

0.082

TOP PERFORMER: HYBRID

COVERAGE

Recall@10

0.075

SINGLE RELEVANT ITEM HIT-RATE

RANKING

MAP@10

0.068

MEAN AVERAGE PRECISION

QUALITY

NDCG@10

0.264

DISCOUNTED GAIN (HYBRID A=0.5)

Comparisons show that Hybrid ranking significantly outperforms individual Content and Popularity baselines in ranking quality (NDCG).

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Thank You

A complete, reproducible architecture for hybrid music discovery.

CONTACT & SOURCE

Prepared for Nirmit Gupta · October 2026

PROJECT WRAP-UP

KEY TAKEAWAYS

- **Scalable:** Sparse computation handles large catalogs.
- **Balanced:** Hybrid α-blend optimizes discovery quality.
- **Resilient:** Graceful degradation to content-based features.

FUTURE ROADMAP

- **Data:** Integrate real listening logs (e.g. LFM-1b).
- **Models:** Matrix Factorization (ALS) & Deep Embedding.
- **Features:** Multi-song seed & user profile support.

PROJECT STATUS

Ready for Production Deployment

→ FINAL GATE: METRICS VALIDATED